

به نام خدا

تمرین سری ۲

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فرزان رحمانی

۴۰۳۲۱۰۷۲۵

سوال ۲۲

- 3.22. Consider an LTI system that is stable and for which $H(z)$, the z -transform of the impulse response, is given by

$$H(z) = \frac{3}{1 + \frac{1}{3}z^{-1}}.$$

Suppose $x[n]$, the input to the system, is a unit step sequence.

- (a) Find the output $y[n]$ by evaluating the discrete convolution of $x[n]$ and $h[n]$.
(b) Find the output $y[n]$ by computing the inverse z -transform of $Y(z)$.

(الف)

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} h[k]x[n-k] \\ &= \sum_{k=-\infty}^{\infty} (3(-\frac{1}{3})^k u[k]) u[n-k] \\ &= \sum_{k=0}^n 3(-\frac{1}{3})^k \\ &= \begin{cases} \frac{9}{4} \left(1 - \left(-\frac{1}{3} \right)^{n+1} \right), & n \geq 0 \\ 0, & \text{در غیر این صورت} \end{cases} \end{aligned}$$

(ب)

$$\begin{aligned} Y(z) &= H(z)X(z) \\ &= \frac{3}{1 + \frac{1}{3}z^{-1}} \frac{1}{1 - z^{-1}} \\ &= \frac{\frac{3}{4}}{1 + \frac{1}{3}z^{-1}} + \frac{\frac{9}{4}}{1 - z^{-1}} \end{aligned}$$

$$\begin{aligned}
 y[n] &= \frac{3}{4} \left(-\frac{1}{3}\right)^n u[n] + \frac{9}{4} u[n] \\
 &= \frac{9}{4} \left(1 + \frac{1}{3} \left(-\frac{1}{3}\right)^n\right) u[n] \\
 &= \frac{9}{4} \left(1 - \left(-\frac{1}{3}\right)^{n+1}\right) u[n]
 \end{aligned}$$

سوال ۲۳

3.23. An LTI system is characterized by the system function

$$H(z) = \frac{(1 - \frac{1}{2}z^{-2})}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}, \quad |z| > \frac{1}{2}.$$

- (a) Determine the impulse response of the system.
 (b) Determine the difference equation relating the system input $x[n]$ and the system output $y[n]$.

(الف)

$$\begin{aligned}
 H(z) &= \frac{1 - \frac{1}{2}z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} \\
 &= -4 + \frac{5 + \frac{7}{2}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} \\
 &= -4 - \frac{2}{1 - \frac{1}{2}z^{-1}} + \frac{7}{1 - \frac{1}{4}z^{-1}} \\
 h[n] &= -4\delta[n] - 2\left(\frac{1}{2}\right)^n u[n] + 7\left(\frac{1}{4}\right)^n u[n]
 \end{aligned}$$

(ب)

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n] - \frac{1}{2}x[n-2]$$

سوال ۲۴

3.24. Sketch each of the following sequences and determine their z -transforms, including the region of convergence:

- (a) $\sum_{k=-\infty}^{\infty} \delta[n-4k]$
 (b) $\frac{1}{2} \left[e^{j\pi n} + \cos\left(\frac{\pi}{2}n\right) + \sin\left(\frac{\pi}{2} + 2\pi n\right) \right] u[n]$

نمودارهای دنباله‌ها در زیر نشان داده شده‌اند.

(الف) فرض کنید

$$a[n] = \sum_{k=-\infty}^{\infty} \delta[n - 4k],$$

آنگاه

$$A(z) = \sum_{k=-\infty}^{\infty} z^{-4k}, \quad \text{ROC is empty set since signal is two-sided}$$

توجه کنید چون سیگنال دو طرفه است پس ناحیه همگرایی آن تهی است تبدیل z آن وجود ندارد.

(ب)

$$b[n] = \frac{1}{2} [e^{j\pi n} + \cos(\frac{\pi}{2}n) + \sin(\frac{\pi}{2} + 2\pi n)] u[n]$$

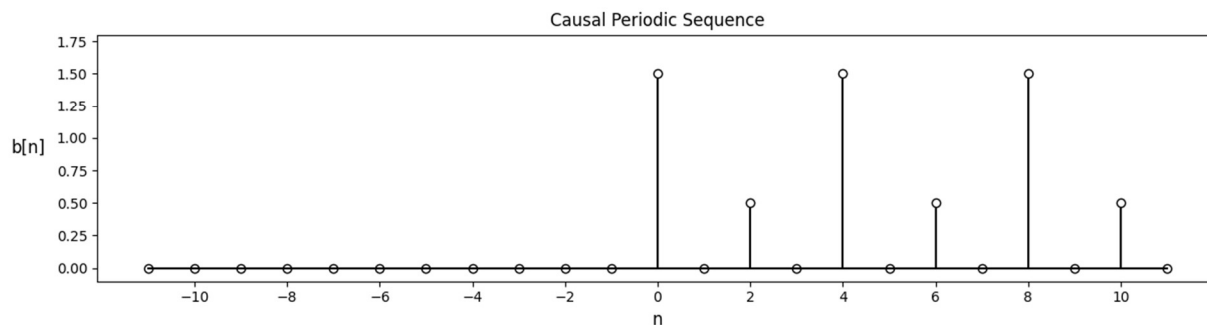
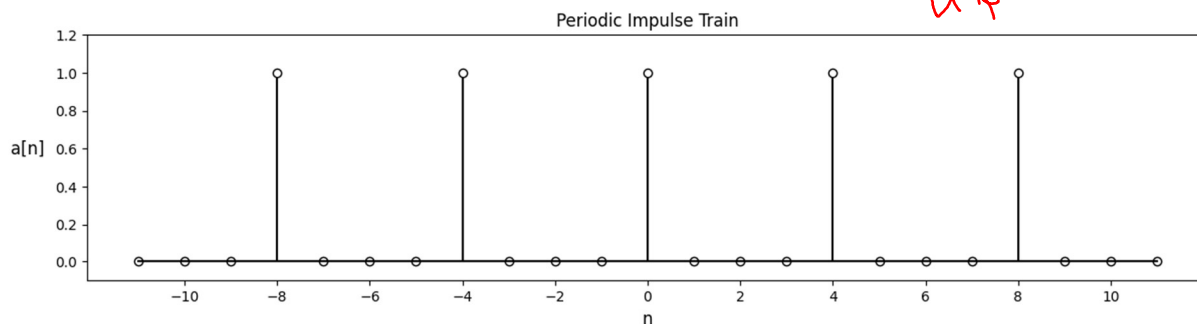
$$= \frac{1}{2} [(-1)^n + \cos(\frac{\pi}{2}n) + 1] u[n]$$

$$= \begin{cases} \frac{3}{2}, & n = 4k, k \geq 0 \\ \frac{1}{2}, & n = 4k + 2, k \geq 0 \\ 0, & \text{در غیر این صورت} \end{cases}$$

$$B(z) = \sum_{n=0}^{\infty} \frac{3}{2} z^{-4n} + \sum_{n=0}^{\infty} \frac{1}{2} z^{-(4n+2)}$$

$$= \frac{3/2 + 1/2 z^{-2}}{1 - z^{-4}}, \quad |z| > 1$$

شبهه
u[n]
up sample



کد پایتون رسم نمودار:

```

import matplotlib.pyplot as plt
import numpy as np

# 1. Define the range for n (discrete time axis)
n = np.arange(-11, 12)

# 2. Define signal a[n]
# This is a periodic impulse train with a period of 4
a = np.zeros_like(n, dtype=float)
for i, val in enumerate(n):
    if val % 4 == 0:
        a[i] = 1.0

# 3. Define signal b[n]
# This appears to be a causal signal (0 for n < 0)
# For n >= 0, it repeats a pattern of [1.5, 0, 0.5, 0]
b = np.zeros_like(n, dtype=float)
for i, val in enumerate(n):
    if val >= 0: # The graph shows 0 for negative values
        if val % 4 == 0:
            b[i] = 1.5
        elif val % 4 == 2:
            b[i] = 0.5

# 4. Create the Plot
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 6))
plt.subplots_adjust(hspace=0.5) # Add space between plots

# Plot a[n]
markerline, stemlines, baseline = ax1.stem(n, a, basefmt="k-")
plt.setp(markerline, markerfacecolor='white', markeredgecolor='black', markersize=6)
plt.setp(stemlines, color='black')
ax1.set_ylabel('a[n]', fontsize=12, rotation=0, labelpad=20)
ax1.set_xlabel('n', fontsize=12)
ax1.set_xticks(np.arange(-10, 12, 2)) # Match ticks to original image
ax1.set_ylim(-0.1, 1.2)
ax1.set_title('Periodic Impulse Train')

# Plot b[n]
markerline, stemlines, baseline = ax2.stem(n, b, basefmt="k-")
plt.setp(markerline, markerfacecolor='white', markeredgecolor='black', markersize=6)
plt.setp(stemlines, color='black')
ax2.set_ylabel('b[n]', fontsize=12, rotation=0, labelpad=20)
ax2.set_xlabel('n', fontsize=12)
ax2.set_xticks(np.arange(-10, 12, 2))
ax2.set_ylim(-0.1, 1.8)
ax2.set_title('Causal Periodic Sequence')
plt.show()

```

3.26. Determine the inverse z -transform of each of the following. In Parts (a)–(c), use the methods specified. In Part (d), use any method you prefer.

(a) Long division:

$$X(z) = \frac{1 - \frac{1}{3}z^{-1}}{1 + \frac{1}{3}z^{-1}}, \quad x[n] \text{ a right-sided sequence}$$

(b) Partial fraction:

$$X(z) = \frac{3}{z - \frac{1}{4} - \frac{1}{8}z^{-1}}, \quad \underline{x[n] \text{ stable}}$$

(c) Power series:

$$X(z) = \ln(1 - 4z), \quad |z| < \frac{1}{4}$$

$$(d) \quad X(z) = \frac{1}{1 - \frac{1}{3}z^{-3}}, \quad |z| > (3)^{-1/3}$$

(الف)

$x[n]$ یک سیگنال دست راستی است و

$$X(z) = \frac{1 - \frac{1}{3}z^{-1}}{1 + \frac{1}{3}z^{-1}}$$

با روش تقسیم طولانی (long division) داریم:

$$\begin{array}{r}
 1 \quad -\frac{2}{3}z^{-1} + \frac{2}{9}z^{-2} + \dots \\
 \hline
 1 + \frac{1}{3}z^{-1} \quad | \quad 1 \quad -\frac{1}{3}z^{-1} \\
 \quad \quad \quad 1 \quad -\frac{1}{3}z^{-1} \\
 \quad \quad \quad \hline
 \quad \quad \quad -\frac{2}{3}z^{-1} \\
 \quad \quad \quad -\frac{2}{3}z^{-1} \quad -\frac{2}{9}z^{-2} \\
 \quad \quad \quad \hline
 \quad \quad \quad \quad \quad \quad +\frac{2}{9}z^{-2}
 \end{array}$$

بنابراین،

$$x[n] = 2\left(-\frac{1}{3}\right)^n u[n] - \delta[n]$$

(ب)

$$X(z) = \frac{3}{z - \frac{1}{4} - \frac{1}{8}z^{-1}} = \frac{3z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})} = \frac{4}{1 - \frac{1}{2}z^{-1}} - \frac{4}{1 - \frac{1}{4}z^{-1}}$$

قطب‌ها در $\frac{1}{2}$ و $-\frac{1}{4}$ هستند. $x[n]$ پایدار است و چون $|z| > \frac{1}{2}$ پس علی است.

بنابراین،

$$x[n] = 4\left(\frac{1}{2}\right)^n u[n] - 4\left(-\frac{1}{4}\right)^n u[n]$$

(७)

$$\begin{aligned} X(z) &= \ln(1-4z) \mid z| < \frac{1}{4} \\ &= - \sum_{i=1}^{\infty} \frac{(4z)^i}{i} = - \sum_{\ell=-\infty}^{-1} \frac{1}{\ell} (4z)^{-\ell} \end{aligned}$$

بنا بر این،

$$x[n] = \frac{1}{n} (4)^{-n} u[-n - 1]$$

(د)

$$X(z) = \frac{1}{1 - \frac{1}{3}z^{-3}} |z| > (3)^{-\frac{1}{3}} \Rightarrow \text{على}$$

با تقسیم طولانی (long division) داریم:

$$\begin{array}{cccc}
 & 1 & +\frac{1}{3}z^{-3} & +\frac{1}{9}z^{-6} & +\dots \\
 & \frac{1}{-} \frac{1}{-} \frac{1}{-} & \frac{1}{-} \frac{1}{-} \frac{1}{-} & \frac{1}{-} \frac{1}{-} \frac{1}{-} & \frac{1}{-} \frac{1}{-} \frac{1}{-} \\
 & 1 & -\frac{1}{3}z^{-3} & & \\
 1-\frac{1}{3}z^{-3} \mid & \frac{1}{-} \frac{1}{-} \frac{1}{-} & \frac{1}{-} \frac{1}{-} \frac{1}{-} & & \\
 & & +\frac{1}{3}z^{-3} & & \\
 & & +\frac{1}{3}z^{-3} & -\frac{1}{9}z^{-6} & \\
 & & \frac{1}{-} \frac{1}{-} \frac{1}{-} & \frac{1}{-} \frac{1}{-} \frac{1}{-} & \\
 & & & +\frac{1}{9}z^{-6} &
 \end{array}$$

$$\Rightarrow x[n] = \begin{cases} \left(\frac{1}{3}\right)^{\frac{n}{3}}, & n = 0, 3, 6, \dots \\ 0, & \text{در غیر این صورت} \end{cases}$$

3.32. The pole-zero diagram in Figure P3.32-1 corresponds to the z-transform $X(z)$ of a causal sequence $x[n]$. Sketch the pole-zero diagram of $Y(z)$, where $y[n] = x[-n+3]$. Also, specify the region of convergence for $Y(z)$.

$$y[n] = x[-(n-3)]$$

$$x[n] \rightarrow x[n+3] \rightarrow x[-n+3]$$

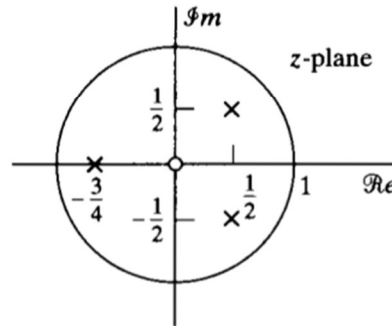


Figure P3.32-1

از نمودار قطب-صفر داریم

$$X(z) = \frac{z}{\left(z^2 - z + \frac{1}{2}\right)\left(z + \frac{3}{4}\right)} \quad |z| > \frac{3}{4}$$

$$y[n] = x[-n+3] = x[-(n-3)]$$

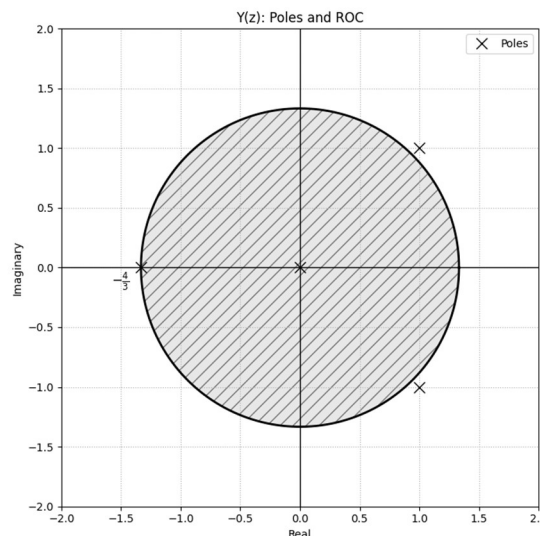
$$\Rightarrow Y(z) = z^{-3}X(z^{-1}) = \frac{z^{-3}z^{-1}}{(z^{-2} - z^{-1} + \frac{1}{2})(z^{-1} + \frac{3}{4})}$$

$$= \frac{8/3}{z(2 - 2z + z^2)(\frac{4}{3} + z)}$$

قطب‌ها در $0, -\frac{4}{3}, 1 \pm j$ هستند، صفر در ∞ است.

$x[n]$ علی است پس $x[-n+3]$ یک سیگنال دست چپ است پس ناحیه همگرایی $0 < |z| < 4/3$ است.

time reversal



```

import numpy as np
import matplotlib.pyplot as plt
# System specifications
poles = np.array([
    0 + 0j,
    -4/3 + 0j,
    1 + 1j,
    1 - 1j
])
roc_radius = 4/3
# Create plot
fig, ax = plt.subplots(figsize=(6, 6))
# --- Plot ROC disk ---
theta = np.linspace(0, 2*np.pi, 400)
x = roc_radius * np.cos(theta)
y = roc_radius * np.sin(theta)
ax.fill(x, y, facecolor='lightgray', alpha=0.5,
        hatch='//', edgecolor='black')
# --- Plot circle boundary ---
ax.plot(x, y, color='black', linewidth=2)
# --- Plot poles ---
ax.plot(poles.real, poles.imag,
        marker='x', markersize=10,
        linestyle='None', color='black',
        label='Poles')
# --- Axes lines ---
ax.axhline(0, color='black', linewidth=1)
ax.axvline(0, color='black', linewidth=1)
# Plot styling
ax.set_aspect('equal', adjustable='box')
limit = 2
ax.set_xlim([-limit, limit])
ax.set_ylim([-limit, limit])
ax.set_xlabel('Real')
ax.set_ylabel('Imaginary')
ax.set_title('Y(z): Poles and ROC')
# Label the -4/3 pole (as in the image)
ax.text(-4/3 - 0.25, -0.15, r'$-\frac{4}{3}$', fontsize=12)
ax.grid(True, linestyle=':')
# Legend
ax.legend(loc='upper right')
plt.show()

```

3.33. Let $x[n]$ be the sequence with the pole-zero plot shown in Figure P3.33-1. Sketch the pole-zero plot for:

(a) $y[n] = \left(\frac{1}{2}\right)^n x[n]$

(b) $w[n] = \cos\left(\frac{\pi n}{2}\right) x[n]$

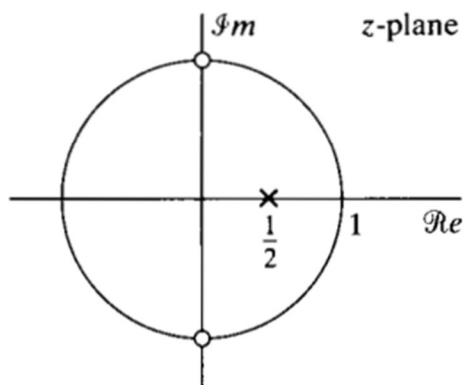


Figure P3.33-1

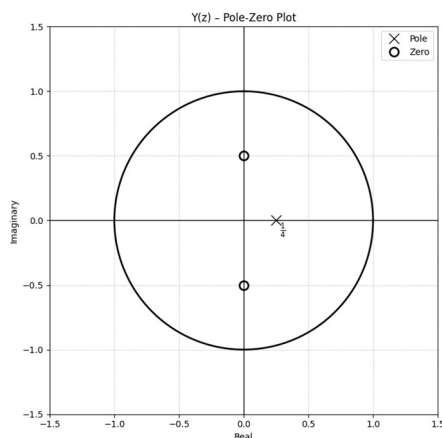
از نمودار قطب-صفر داریم

$$X(z) = \frac{z^2 + 1}{z - \frac{1}{2}}$$

(الف)

$$y[n] = \left(\frac{1}{2}\right)^n x[n] \Rightarrow Y(z) = X(2z) = \frac{4z^2 + 1}{2z - \frac{1}{2}}$$

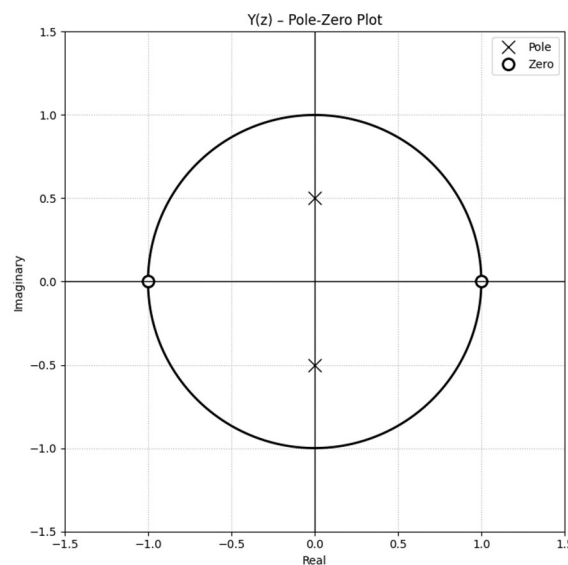
صفرها: $\pm \frac{1}{2}j$
 قطبها: $\frac{1}{4}, \infty$



(ب)

$$w[n] = \cos\left(\frac{\pi n}{2}\right)x[n] = \frac{1}{2}(e^{j\pi n/2} + e^{-j\pi n/2})x[n]$$
$$W(z) = \frac{1}{2}X(e^{-j\pi/2}z) + \frac{1}{2}X(e^{j\pi/2}z) = \frac{1}{2}X(-jz) + \frac{1}{2}X(jz)$$
$$W(z) = \frac{1}{2}\left(\frac{-z^2 + 1}{-jz - \frac{1}{2}}\right) + \frac{1}{2}\left(\frac{-z^2 + 1}{jz - \frac{1}{2}}\right) = \frac{z^2 - 1}{2(z^2 + \frac{1}{4})}$$

قطب‌ها در $\pm \frac{1}{2}j$ هستند.
صفرها در ± 1 هستند.



سوال ۳۶

3.36. If the input $x[n]$ to an LTI system is $x[n] = u[n]$, the output is

$$y[n] = \left(\frac{1}{2}\right)^{n-1} u[n+1].$$

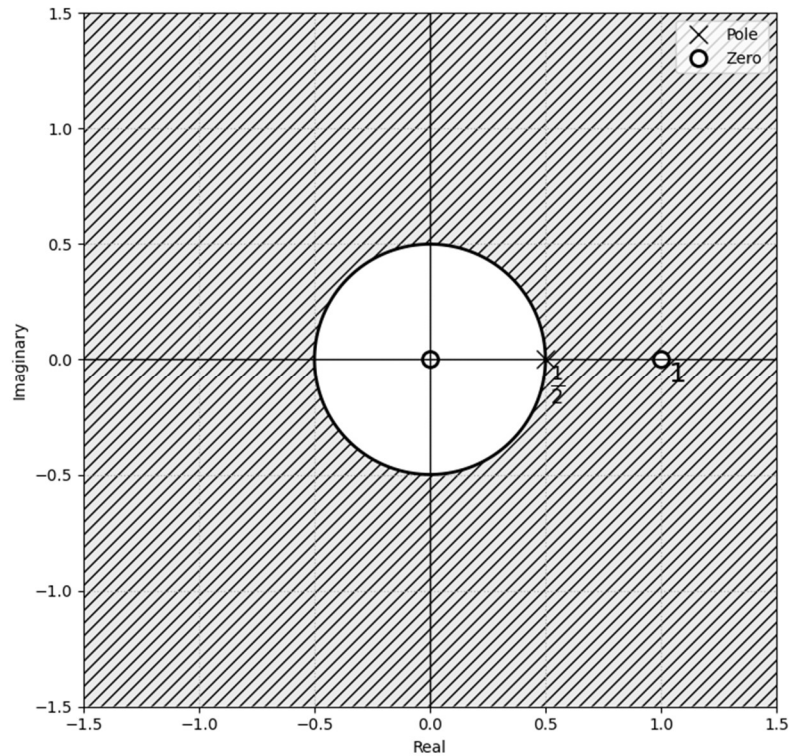
- (a) Find $H(z)$, the z -transform of the system impulse response, and plot its pole-zero diagram.
- (b) Find the impulse response $h[n]$.
- (c) Is the system stable?
- (d) Is the system causal?

$$x[n] = u[n] \Leftrightarrow X(z) = \frac{1}{1 - z^{-1}} \quad |z| > 1$$
$$y[n] = \left(\frac{1}{2}\right)^{n-1} u[n+1] = 4 \left(\frac{1}{2}\right)^{n+1} u[n+1] \Leftrightarrow Y(z) = \frac{4z}{1 - \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2}$$

Shift: $n+1$

(الف)

$$H(z) = \frac{Y(z)}{X(z)} = \frac{4z(1 - z^{-1})}{1 - \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2}$$



(ب)

partial
fraction

$$H(z) = \frac{4z}{1 - \frac{1}{2}z^{-1}} - \frac{4}{1 - \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2}$$

$$h[n] = 4\left(\frac{1}{2}\right)^{n+1}u[n+1] - 4\left(\frac{1}{2}\right)^n u[n]$$

$$= 4\underline{\delta[n+1]} - 2\left(\frac{1}{2}\right)^n u[n]$$

(ج) ناحیه همگرایی $H(z)$ شامل $|z| = 1$ می شود پس پایدار است.

(د) از قسمت (ب) می بینیم که $h[n]$ از $n = -1$ شروع می شود پس علی نیست. دست راستی دهست.

causal $\rightarrow 0$ for $n < 0$

3.40. In Figure P3.40-1, $H(z)$ is the system function of a causal LTI system

- (a) Using z -transforms of the signals shown in the figure, obtain an expression for $W(z)$ in the form

$$W(z) = H_1(z)X(z) + H_2(z)E(z),$$

where both $H_1(z)$ and $H_2(z)$ are expressed in terms of $H(z)$.

- (b) For the special case $H(z) = z^{-1}/(1 - z^{-1})$, determine $H_1(z)$ and $H_2(z)$.
 (c) Is the system $H(z)$ stable? Are the systems $H_1(z)$ and $H_2(z)$ stable?

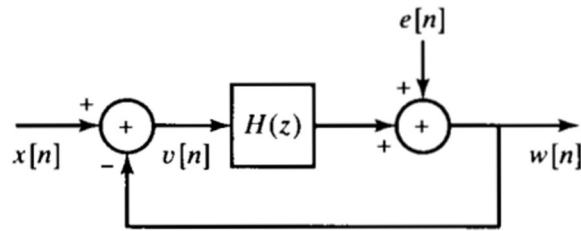


Figure P3.40-1

(الف) پس از نوشتن تساوی‌های زیر:

$$V(z) = X(z) - W(z)$$

$$W(z) = V(z)H(z) + E(z)$$

برای $W(z)$ معادلات را حل می‌کنیم:

$$W(z) = \frac{H(z)}{1 + H(z)}X(z) + \frac{1}{1 + H(z)}E(z)$$

(ب)

$$H_1(z) = \frac{H(z)}{1 + H(z)} = \frac{\frac{z^{-1}}{1 - z^{-1}}}{1 + \frac{z^{-1}}{1 - z^{-1}}} = z^{-1}$$

$$H_2(z) = \frac{1}{1 + \frac{z^{-1}}{1 - z^{-1}}} = 1 - z^{-1}$$

(ج) $H(z)$ به دلیل قطبش در $z = 1$ پایدار نیست، اما $H_1(z)$ و $H_2(z)$ پایدار هستند چرا که در $|z| = 1$ قطب ندارند.

- 3.46. The signal $y[n]$ is the output of an LTI system with impulse response $h[n]$ for a given input $x[n]$. Throughout the problem, assume that $y[n]$ is stable and has a z -transform $Y(z)$ with the pole-zero diagram shown in Figure P3.46-1. The signal $x[n]$ is stable and has the pole-zero diagram shown in Figure P3.46-2.

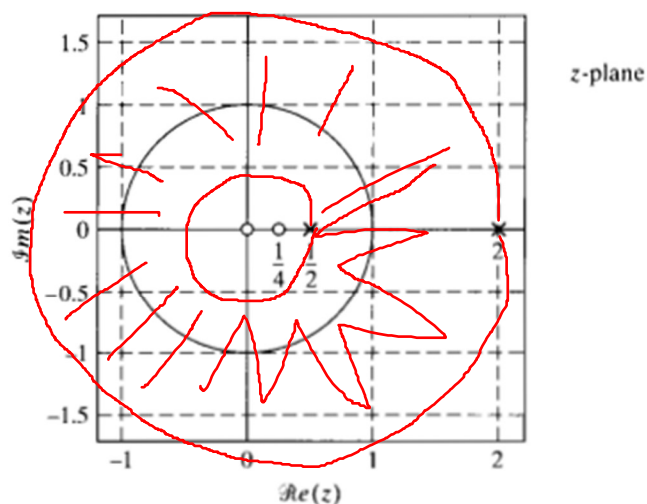


Figure P3.46-1

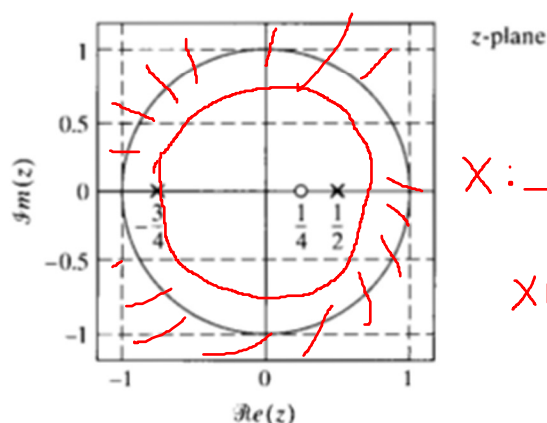


Figure P3.46-2

- What is the region of convergence, $Y(z)$?
- Is $y[n]$ left sided, right sided, or two sided?
- What is the ROC of $X(z)$?
- Is $x[n]$ a causal sequence? That is, does $x[n] = 0$ for $n < 0$?
- What is $x[0]$?
- Draw the pole-zero plot of $H(z)$, and specify its ROC.
- Is $h[n]$ anticausal? That is, does $h[n] = 0$ for $n > 0$?

(الف) از آنجا که $y[n]$ پایدار است، ناحیه همگرایی آن شامل دایره واحد می‌شود. بنابراین، $Y(z)$ برای $|z| < 2$ همگرا است.

(ب) از آنجا که ناحیه همگرایی یک حلقه در صفحه z است، $y[n]$ یک دنباله دوطرفه (two-sided) است.

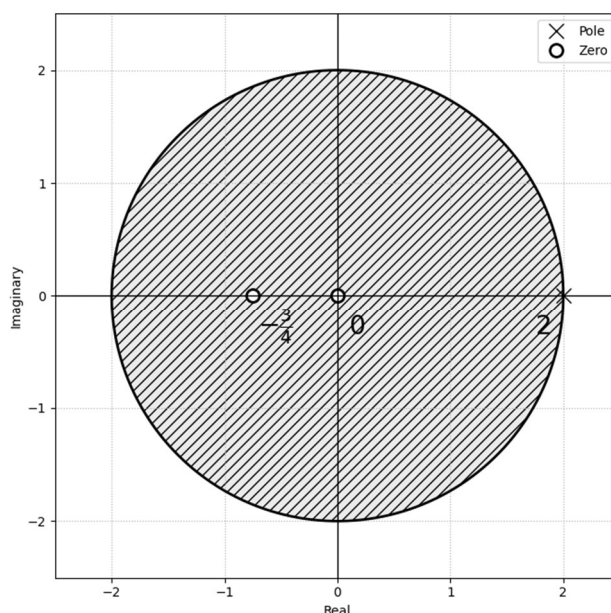
(ج) $x[n]$ پایدار است، بنابراین ناحیه همگرایی آن شامل دایره واحد است. همچنین، یک صفر در بینهایت دارد، بنابراین ناحیه همگرایی شامل بینهایت می‌شود. ناحیه همگرایی: $|z| > \frac{3}{4}$.

(د) از آنجا که ناحیه همگرایی $x[n]$ شامل بینهایت است، $X(z)$ شامل هیچ توان مثبتی از z نیست، و بنابراین $x[n] = 0$ برای $n < 0$. در نتیجه $x[n]$ علی است.

(ه)

$$x[0] = X(z) \big|_{z=\infty} = \frac{A(1 - \frac{1}{4}z^{-1})}{(1 + \frac{3}{4}z^{-1})(1 - \frac{1}{2}z^{-1})} \big|_{z=\infty} = \frac{A \cdot \cancel{\infty}}{\infty \cdot \cancel{\infty}} = \frac{A}{\infty} = 0$$

(و) $H(z)$ صفرهایی در $-\frac{3}{4}$ و 0 ، و قطب‌هایی در 2 و ∞ دارد. ناحیه همگرایی آن $|z| < 2$ است.



(ز) از آنجا که ناحیه همگرایی $h[n]$ شامل 0 است، $H(z)$ شامل هیچ توان منفی از z نیست، که دلالت بر $h[n] = 0$ برای $n > 0$ می‌کند. بنابراین $h[n]$ ضدعلی (anti-causal) است.

3.53. A real finite-duration sequence whose z -transform has no zeros at conjugate reciprocal pair locations and no zeros on the unit circle is uniquely specified to within a positive scale factor by its Fourier transform phase (Hayes et al., 1980).

An example of zeros at conjugate reciprocal pair locations is $z = a$ and $(a^*)^{-1}$. Even though we can generate sequences that do not satisfy the preceding set of conditions, almost any sequence of practical interest satisfies the conditions and therefore is uniquely specified to within a positive scale factor by the phase of its Fourier transform.

Consider a sequence $x[n]$ that is real, that is zero outside $0 \leq n \leq N-1$, and whose z -transform has no zeros at conjugate reciprocal pair locations and no zeros on the unit circle. We wish to develop an algorithm that reconstructs $cx[n]$ from $\angle X(e^{j\omega})$, the Fourier transform phase of $x[n]$, where c is a positive scale factor.

(a) Specify a set of $(N-1)$ linear equations, the solution to which will provide the recovery of $x[n]$ to within a positive or negative scale factor from $\tan\{\angle X(e^{j\omega})\}$. You do not have to prove that the set of $(N-1)$ linear equations has a unique solution. Further, show that if we know $\angle X(e^{j\omega})$ rather than just $\tan\{\angle X(e^{j\omega})\}$, the sign of the scale factor can also be determined.

(b) Suppose

$$x[n] = \begin{cases} 0, & n < 0, \\ 1, & n = 0, \\ 2, & n = 1, \\ 3, & n = 2, \\ 0, & n \geq 3. \end{cases}$$

Using the approach developed in Part (a), demonstrate that $cx[n]$ can be determined from $\angle X(e^{j\omega})$, where c is a positive scale factor.

$$\theta_z(\omega) = \tan^{-1} \left(\frac{\text{Im}\{X(e^{j\omega})\}}{\text{Re}\{X(e^{j\omega})\}} \right) \Rightarrow \tan \theta_z(\omega) = \frac{-\sum_{n=0}^{N-1} x[n] \sin(n\omega)}{\sum_{n=0}^{N-1} x[n] \cos(n\omega)}$$

$$\tan \theta_z(\omega) \sum_{n=0}^{N-1} x[n] \cos(n\omega) = - \sum_{n=1}^{N-1} x[n] \sin(n\omega)$$

$$\tan \theta_z(\omega) x[0] + \sum_{n=1}^{N-1} x[n] (\tan \theta_z(\omega) \cos(n\omega) + \sin(n\omega)) = 0$$

$$\tan \theta_z(\omega_k) + \frac{1}{x[0]} \sum_{n=1}^{N-1} x[n] (\tan \theta_z(\omega_k) \cos(n\omega_k) + \sin(n\omega_k)) = 0$$

برای $N-1$ مقدار از ω_k در بازه $0 < \omega_k < \pi$

(ب)

$$x[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] \Rightarrow X(z) = 1 + 2z^{-1} + 3z^{-2}$$

$$\theta_z(\omega) = \tan^{-1} \left(\frac{-2\sin(\omega) - 3\sin(2\omega)}{1 + 2\cos(\omega) + 3\cos(2\omega)} \right)$$

مقادیر $\theta_z(\frac{\pi}{2}) = \frac{5\pi}{4}$ و $\theta_z(\frac{2\pi}{3}) = \frac{5\pi}{6}$ را در نظر بگیرید که معادلات زیر را می‌دهند:

$$\tan \theta_x \left(\frac{\pi}{2} \right) + \frac{1}{x[0]} \left[x[1] \left(\tan \theta_x \left(\frac{\pi}{2} \right) \cos \frac{\pi}{2} + \sin \frac{\pi}{2} \right) + x[2] \left(\tan \theta_x \left(\frac{\pi}{2} \right) \cos \pi + \sin \pi \right) \right] = 0$$

$$\tan \theta_x \left(\frac{2\pi}{3} \right) + \frac{1}{x_0} \left[x[1] \left(\tan \theta_x \left(\frac{2\pi}{3} \right) \cos \frac{2\pi}{3} + \sin \frac{2\pi}{3} \right) + x[2] \left(\tan \theta_x \left(\frac{2\pi}{3} \right) \cos \frac{4\pi}{3} + \sin \frac{4\pi}{3} \right) \right] = 0$$

$$1 + \frac{1}{x[0]} (x[1] \cdot 1 + x[2] \cdot -1) = 0$$

$$-\frac{1}{\sqrt{3}} + \frac{1}{x[0]} \left(x[1] + \frac{1}{2\sqrt{3}} + \frac{\sqrt{3}}{2} \right) + x[2] \left(\frac{1}{2\sqrt{3}} - \frac{\sqrt{3}}{2} \right) = 0$$

$$\begin{cases} x[0] + x[1] - x[2] = 0 \\ -x[0] + 2x[1] - x[2] = 0 \end{cases} \Rightarrow \begin{cases} x[1] = 2x[0] \\ x[2] = 3x[0] \end{cases}$$

بنابراین

$$x[n] = x[0](\delta[n] + 2\delta[n-1] + 3\delta[n-2])$$

که در آن $x[0]$ نامعین است.

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